Short Read Aligners

Alignments are subjective measures

Different algorithms and scoring matrices may produce entirely different outcomes.

There is always that unconfortable uncertainty:

- *What is "reality" and what is caused by your choice of algorithm, method, scoring?*
- *Would I get something completely different if I used different parameters?*

Sequencing as "measurement"

Wouldn't it be nice if sequencing worked like other scientific **measurements**?

Where we could repeatedly measure the same thing over and over, and with that assess the error?

This is what high throughput sequencing was designed to do.

High throughput sequencing

Millions of DNA strands broken into short fragments.

```
==================================
==================================
==================================
```

For example, one outcome could be:

```
=== ===== ==== === = = ===== === ===
== ==== ===== === === = ====== == ==
======= == == = = = ===== === ======
```

Each region is "covered" by multiple short fragments.
We sequence a subset of these pieces.

The promise of high-throughput sequencing

Since each fragment is short, most of the time it won't be all that different from the reference.

Aligning **short, very similar** sequences is **far less sensitive** to parameters than aligning **very different, long** sequences.

Since we can also measure the same region many times, over and over, hopefully that ought to help us identify errors ...

High troughput sequencing applications

1. Resequencing

Compare sequences to a known **reference**.

2. Quantification

Assess the abundance of the sequenced fragments.

3. Assembly

Build full sequences out of short fragments.

4. Classification

Identify the taxonomical rank of sequences.

Sanger Sequencing

Old school up to 2005: Sanger instrumentation produces few (thousands), but long reads (`1000bp`). These are suited for:

- Finding local similarities -> homology

Short Read Sequencing

New school, since 2005: High throughput sequencing produces a vast number (billions) of very short reads (`200bp`) that are suited for:

- Finding differences relative to known information.
- Quantifying abundance of various fragments.

Long Read Sequencing Even

An even **Newer school, since 2015** - PacBio, MinION. Much longer reads, up to 20Kb or perhaps longer.

- Very error prone though!
- Lots of "systematic" errors.

New techniques to reduce errors by re-resequencing the same DNA fragment within the same sequencing process.

Instead of **1 x 20Kb** --> **3 x 6kb** (consensus calling) - reads the fragment 3 times (PacBio).

Personal Opinion

Consensus Calling

The most impactful development since 2005

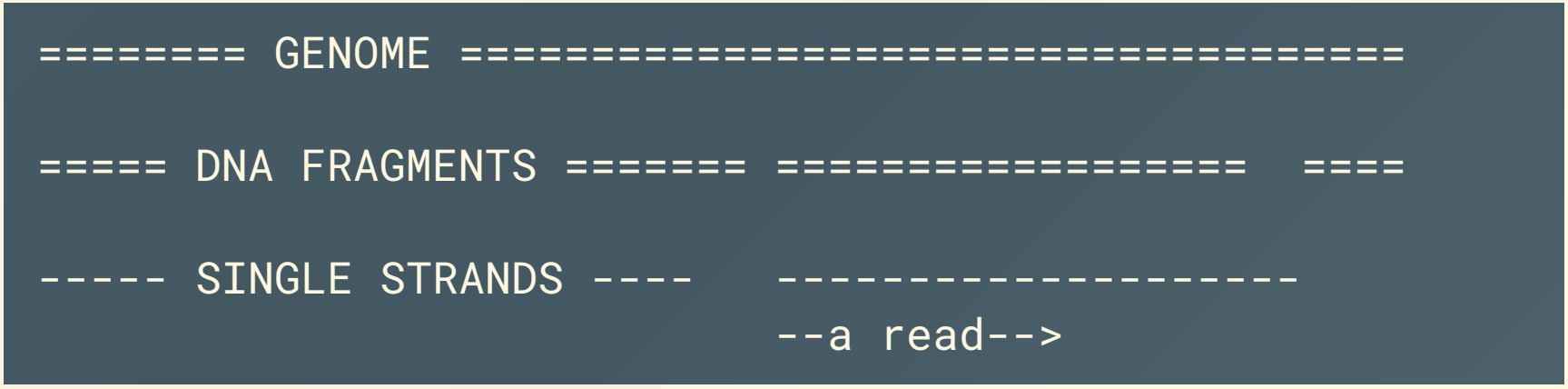
Whole chapters of science will need to be redone with consensus calling.

What is a sequencing read?

One read == one (short) measurement

We use the word read a lot

Make sure you understand what it is.



A read is a partial sequence measurement that originates from a strand of a DNA fragment observed in the DNA library.

Typically the read will start at the 5' end of the single stranded DNA.

How do short read aligners work?

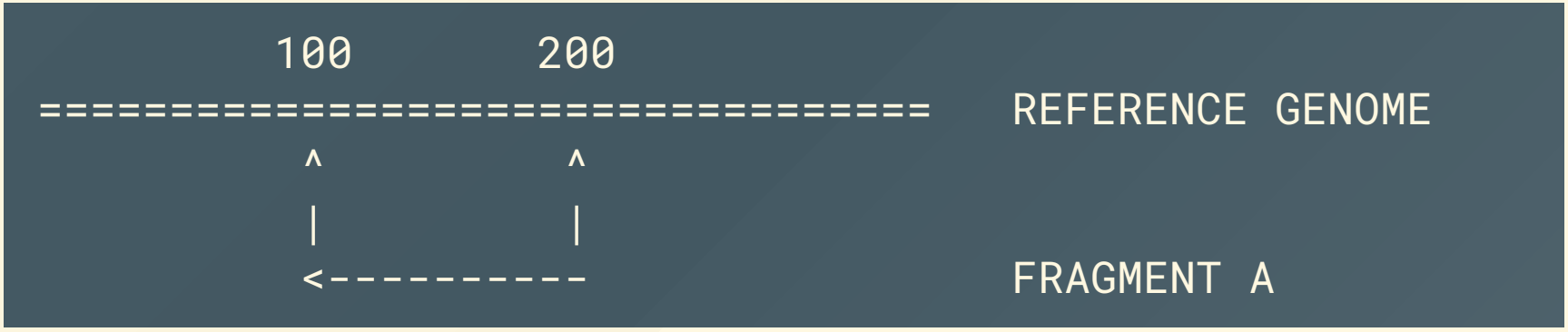
Use heuristics to quickly identify locations (hits) where the reads matches **well**.

Cannot identify locations where reads **do not match** all that well.

Can only see **high scoring** alignments and won't find **low scoring** alignments.

What is a mapping?

A mapping means finding the location where the sequenced DNA fragment originated from.



The mapping here could be reported as:

- `Fragment A` maps to the region between `100` and `200` on the reverse strand of the `REFERENCE GENOME`

Mapping vs Alignment

There is conceptual difference:

- A mapping is the location where a read sequence is placed.
- A mapping is regarded to be correct if it indicates the true region.

- An alignment is the detailed placement of each base in a read.
- An alignment is regarded to be correct if each base is placed correctly.

Mapping vs Alignment Paradox

It is possible to have correct mapping and incorrect alignment

It is possible to have correct alignment and incorrect mapping.

1. **Variation** studies , **SNP calling** require **accurate alignments**.
2. **RNA-Seq**, **ChIP-Seq** studies require **accurate mapping**.

Short Read Alignment software

Too many to list.

Each claims to be better than the other.

The vast majority of published tools comparisons are unhelpful. It is surprisingly difficult to compare them.

Tool A is faster than **Tool B** and it is implied that the results are the same under all circumstances.

Tool A is more â€˜accurateâ€™ than **Tool B** and we assume the resource utilization is approximately the same under all circumstances.

Short Read Aligners: Problems and limitations

- Some cannot handle indels (insertion/deletions)
- How "bad" is the worse alignment they can find?
- Some can only find nearly exact matches and report all others as "unmapped."
- There may be massive differences in the additional information that report.
- Sometimes you need to pick a tool based on this additional information that may or may not be present.

Short read aligners

We recommend: **bwa**, **bowtie2**, **soap2**

There is no single best tool, the issues to consider:

- documentation can we figure out how it works
- input features what type of input can it handle
- reporting features will it produce the type of output that we can use
- performance is it feasible to run on my resources

So what aligner should I use?

One is that is reasonably correct and can operate with your resources

Results should hold across methods.

If in doubt use:

```
bwa mem
```

Plus it is the only short read aligner that can also produce good long read alignments (with limitations).

How do most short read aligners work?

1. Index the genome, this only needs to be done once

2. Generate the alignment.

We will demonstrate the use of two different short read aligners:

- **bwa**
- **bowtie2**

Google for their homepages/documentation.

Building an index

Get a reference genome (1976 Ebola outbreak):

```
mkdir -p db
efetch -db=nuccore -format=fasta -id=AF086833 > db/1976.fa
```

Build a **bowtie2** index:

```
bowtie2-build db/1976.fa db/1976.fa
```

Build a **bwa** index:

```
bwa index db/1976.fa
```

Both of these commands need to be run just once.

Scripting vs typing at the command line

When using the command line, we typically type out the whole filename.

The `TAB` completion helps to fill in the names.

Never type file names yourself. Start the name then press `TAB`. It is amazingly helpful.

But when you write scripts you should **always** use variables.

It allows you to reuse the same script for a different purpose later. It is a component of **reproducibility**!

In your scripts use variable names

Using variable names allow you to separate the changeable and the identical parts of the calls

```
# Accession number.
ACC=AF086833

# Reference filename.
REF=ref/AF086833.fa

# Directory to hold the references in.
mkdir -p ref

# Fetch the reference genome.
efetch -db=nuccore -format=fasta -id=$ACC > $REF
```

Building indices

Typically the tools asks you to provide an input name and an output base name:

Build a **bowtie2** index:

```
bowtie2-build $REF $REF
```

Build a **bwa** index (will automatically create the output name):

```
bwa index $REF
```

Get a sequencing run

Obtain a sequencing run for the Ebola project:

```
fastq-dump -X 10000 --split-files SRR1972739
```

We note that it is a paired end run:

```
SRR1972739_1.fastq
SRR1972739_2.fastq
```

In a real project we would apply a QC step here if needed, then run the alignment on the improved dataset.

Single end alignment with bowtie

```
R1=SRR1972739_1.fastq
```

Align with **bowtie2**:

```
bowtie2 -x $REF -U $R1 > bowtie2.sam
```

Prints:

```
10000 reads; of these:
  10000 (100.00%) were unpaired; of these:
    3628 (36.28%) aligned 0 times
    6372 (63.72%) aligned exactly 1 time
    0 (0.00%) aligned >1 times
63.72% overall alignment rate
```

Single end alignment with bwa

```
R1=SRR1972739_1.fastq
```

Align with **bwa**:

```
bwa mem $REF $R1 > bwa.sam
```

See statistics on the run:

```
samtools flagstat bwa.sam
```

The output file is in SAM format. We'll cover this format in a different lecture.

Paired-end alignment with bowtie2

```
R1=SRR1972739_1.fastq
R2=SRR1972739_2.fastq
```

Align paired end reads with **bowtie2**:

```
bowtie2 -x $REF -1 $R1 -2 $R2 > bowtie2.sam
```

Paired-end alignment with bwa

```
R1=SRR1972739_1.fastq
R2=SRR1972739_2.fastq
```

Align paired end reads with **bwa**:

```
bwa mem $REF $R1 $R2 > bwa.sam
```

Short read aligner may be tuned as well

There may be a wealth of options as well:

```
bowtie2
```

Prints:

```
 Alignment:
  -N <int>            max # mismatches in seed alignment; c
  -L <int>            length of seed substrings; must be >3
  -i <func>           interval between seed substrings w/r/
  --n-ceil <func>     func for max # non-A/C/G/Ts permitted
  --dpad <int>        include <int> extra ref chars on side
  --gbar <int>        disallow gaps within <int> nucs of re
  --ignore-quals      treat all quality values as 30 on Phr
  --nofw              do not align forward (original) versi
```

Improving alignment/mapping rates

If alignment rates are low:

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1. Try to improve dataset (QC)

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2. Try to tune the aligner with different parameters.

Compare the **bwa** and **bowtie2** help pages.

```
bwa mem

bowtie2
```